Now, as we saw for k-means, when we count on random methods, we can be lucky, but more often than not we won't be: figure 16.3 shows a particularly unlucky assignment for the K_5 graph – as we know from chapter 15, it's possible to draw this complete graph in the plane with a single intersection between its edges.

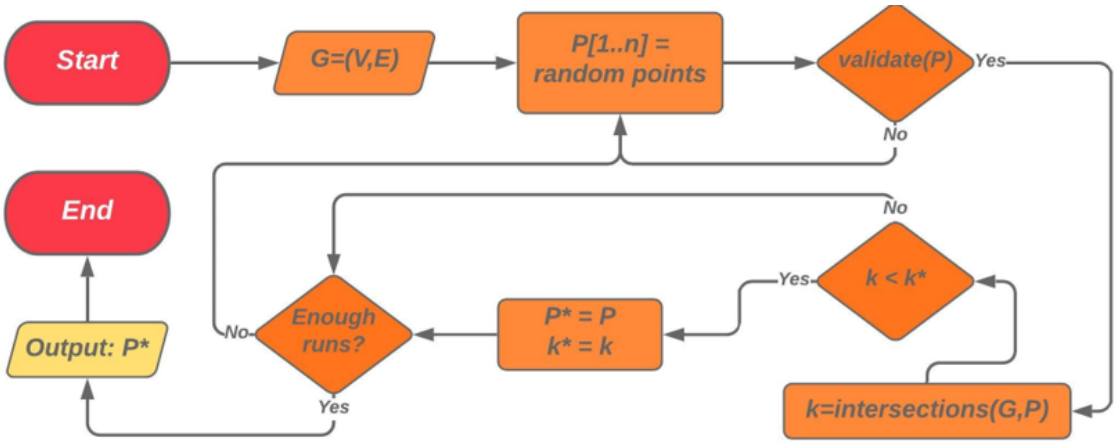


Figure 16.4 A flow chart for the algorithm generating random embeddings and selecting the one with the fewest edge intersections.

A way to raise our chances with k-means was using the random-restart technique, basically running the algorithm (and in turn the random initialization) several times, and then saving the best solution found across all those runs.

This strategy looks interesting for our problem as well:

1. Randomly generate the positions of the vertices;
2. Check that the assignment abides by the constraints for edges and vertices (and discard the current assignments if it doesn't);
3. Compute the number of edge intersections;
4. If it's the best result so far, keep it, otherwise discard it;
5. Restart from 1.